

Case Study - AI Solutions Developer

Background

An insurance company receives hundreds of emails daily from agents and brokers submitting Commercial Auto Insurance applications.

Each email may contain multiple attachments in varying formats, including:

- PDF documents
- Excel spreadsheets
- Images
- Scanned forms
- Photos of driver's licenses
- Motor Vehicle Records (MVRs)
- Loss runs
- Additional supporting documents

The company wants to build an AI-powered system that automates the intake and review of these submissions before they reach an underwriter.

Business Requirement

Design a system that allows agents and brokers to submit insurance applications via email.

The system should:

1. Receive emails and attachments.
2. Process and classify all submitted documents.
3. Extract relevant information from the submission package.
4. Analyze the submission using AI, let AI use industry knowledge and give information about the submission.
5. Apply business rules and underwriting guidelines.
6. Present the final results to users through a web application.

Additional Considerations

The submission packages are often imperfect.

Examples include:

Incorrectly Named Documents

Documents may have misleading or inaccurate file names.

Examples:

- A file named "Drivers.xlsx" may actually contain vehicle information.
- A file named "LossRuns.pdf" may contain driver information.

The system should not rely solely on file names to determine document type.

Multiple Document Formats

Documents may arrive in many different formats and quality levels.

Examples:

- Digitally generated PDFs
- PDFs created from photographs taken on mobile phones
- Scanned documents
- Images
- Handwritten notes

Duplicate Information

The same information may appear in multiple documents.

Conflicting Information

Information may not be consistent across documents.

Examples:

- Vehicle counts may differ between forms.
- Driver information may differ between a Driver's License and an MVR.
- Business information may differ across application documents.
- Dates, VINs, addresses, or policy information may not match.

Example Workflow

An email arrives containing several attachments.

The system should:

Intake

- Receive and store the email and attachments.
- Determine what documents have been submitted.

Document Understanding

- Identify document types.
- Extract relevant information.
- Associate related information across documents.
- Identify missing information.
- Detect duplicate and conflicting information.

AI Processing

Use AI to:

- Summarize the submission.
- Identify potential underwriting concerns.
- Detect inconsistencies across documents.
- Generate recommendations for the underwriting team.
- Assign confidence levels to extracted information and recommendations.

Rules Processing

Apply business rules and underwriting guidelines to determine the appropriate next step.

Examples may include:

- Flagging incomplete submissions.
- Routing submissions to specific underwriting teams.
- Identifying prohibited risks.
- Escalating submissions requiring manual review.

Results

Present a consolidated view containing:

- Submission summary
- Extracted information
- Missing information
- Identified conflicts
- AI findings
- Rule outcomes
- Recommended next actions

Deliverables

Please prepare the following during the interview:

Solution Architecture

Be prepared to discuss:

- Email ingestion
- Document processing
- AI services
- Business rules
- Data storage
- User interface
- Workflow orchestration

AI Design

Explain:

- How documents will be classified
- How information will be extracted
- How confidence levels will be assigned

- How conflicting information will be handled
- How human review will be incorporated

Rules & Workflow Design

Describe:

- How business rules are maintained
- How AI outputs interact with business rules
- How exceptions are handled
- How auditability is maintained

Scalability & Reliability

Explain how your solution would operate at production scale.

Security & Compliance

Describe your approach to:

- Protecting sensitive information
- Access control
- Data encryption
- Audit logging
- Regulatory compliance

User Experience

Describe the workflow and user experience for underwriters and operations teams.

Candidate Instructions

This exercise is designed to evaluate both your ability to design AI-powered solutions and your ability to identify and solve real-world business problems.

During the interview, you will be expected to:

- Explain your proposed solution at a high level.
- Walk through your architecture and design decisions.

- Identify assumptions, risks, and tradeoffs.
- Ask clarifying questions to better understand the business requirements.
- Explain how your solution would handle real-world challenges such as poor-quality documents, missing information, conflicting data, and scalability concerns.

Not all requirements, constraints, business rules, and success criteria have been provided intentionally. Part of the exercise is determining what additional information is required and asking the appropriate questions before designing a solution.

Take-Home Assignment

Following the interview, you will build a working prototype demonstrating your proposed solution.

You may use any programming language, framework, AI models, cloud services, workflow tools, or technologies of your choice.

The prototype should demonstrate the end-to-end workflow, including:

- Email intake or simulated email intake
- Processing multiple attachment types
- Document classification
- Information extraction
- Handling photographed PDFs and scanned documents
- Detection of conflicting information
- AI-powered analysis and recommendations
- Business rule execution
- Presentation of final results

Final Submission

Please submit a video recording demonstrating your working solution.

The video should be approximately 5–15 minutes long and should include:

- A live demonstration of the system
- An explanation of the overall architecture
- Your design decisions and assumptions
- The complete end-to-end workflow
- How the system handles document classification
- How the system extracts information
- How the system handles poor-quality documents
- How the system identifies conflicting information
- How AI recommendations are generated
- How business rules are applied
- How final results are presented to users

We are primarily interested in understanding your thought process, technical approach, and ability to build practical AI solutions. The video should clearly demonstrate a working system and explain how the product operates from end to end.

Timeline

The completed demo video must be submitted no later than **5 calendar days after the interview.**